Good Road - Citizen-Powered Traffic Violation Reporting System

24-26J-265

Project Proposal Report

Rathnayake W.K.D

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology
Sri Lanka

July 2025

Hybrid ANPR System for Robust Number Plate Detection Under Diverse Environmental Challenges

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Declaration page of the candidates & supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor
(Dr. Mahima Weerasinghe)

Date: 29th August 2024

Abstract

Automatic Number Plate Recognition (ANPR) systems are critical components of intelligent transportation systems, enabling vehicle identification for traffic management, law enforcement, and security applications. However, existing deep learning approaches, including YOLO variants and CNN architectures, often exhibit degraded performance in challenging scenarios characterized by fast-moving vehicles, low-light conditions, motion blur, and adverse environmental variations. These limitations are particularly pronounced in non-real-time applications where video data is processed post-acquisition, requiring robust solutions capable of handling diverse quality conditions without temporal constraints.

This research proposes a novel hybrid deep learning framework that synergistically integrates state-of-the-art object detection models (YOLOv9 and YOLOv8n) with advanced optical character recognition engines, specifically EasyOCR, augmented by specialized preprocessing pipelines for motion deblurring and low-light enhancement. The framework incorporates adaptive image quality assessment modules that automatically select optimal preprocessing strategies based on detected degradation patterns, including motion blur estimation, illumination analysis, and noise characterization.

The hybrid architecture employs ensemble learning with adaptive fusion mechanisms that dynamically weight detection confidence scores and character recognition outputs based on image quality metrics and contextual information. A multi-stage refinement process validates recognition results through geometric constraints, character sequence patterns, and regional formatting rules to minimize false positives. The system is designed for batch processing of uploaded video content, enabling comprehensive analysis without real-time processing constraints.

Experimental validation on comprehensive datasets comprising over 20,000 vehicle images captured under varying conditions demonstrates significant performance improvements, achieving a mean Average Precision (mAP) exceeding 98% for number plate detection and recognition accuracy above 92% across diverse scenarios, including

nighttime operations, adverse weather conditions, high-speed vehicle capture, and partial occlusions. Comparative analysis reveals substantial improvements over standalone models and existing hybrid approaches, particularly in challenging environments where traditional systems fail.

The primary contribution lies in developing a robust, scalable ANPR component specifically optimized for post-processing video uploads, addressing critical gaps in current systems through enhanced generalization capabilities and improved accuracy for moving vehicles in adverse environments. This work facilitates practical deployment in smart city infrastructure, mining industry applications, and security monitoring systems where reliable vehicle identification is paramount regardless of capture condititions.

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List of Abbreviations

Abbreviation	Description
ANPR	Automatic Number Plate Recognition
YOLO	You Only Look Once
LPDGAN	License Plate Deblurring Generative Adversarial Network
CNN	Convolutional Neural Network
ITS	Intelligent Transportation Systems
SSD	Single Shot MultiBox Detector

1 Introduction

1.1 Overall Introduction

The rapid urbanization and exponential growth in vehicular traffic have necessitated the development of advanced intelligent transportation systems (ITS) to ensure efficient traffic management, enhanced road safety, and robust security measures. Among these technologies, Automatic Number Plate Recognition (ANPR), also known as Automatic License Plate Recognition (ALPR), stands out as a cornerstone for vehicle identification. ANPR systems utilize computer vision and machine learning algorithms to detect, localize, and recognize vehicle number plates from images or video streams, enabling applications such as toll collection, parking management, stolen vehicle tracking, and traffic violation enforcement [12][13]. In high-stakes environments like highways, urban intersections, and industrial sites (e.g., mining operations), where vehicles often move at high speeds, the ability to accurately process number plates from uploaded video content is crucial for post-incident analysis and comprehensive monitoring systems.

However, traditional ANPR methods, reliant on rule-based image processing techniques like edge detection and morphological operations, struggle with variability in plate designs, lighting conditions, and motion artifacts, leading to suboptimal performance. The challenges are particularly pronounced when processing video uploads containing fast-moving vehicles under adverse conditions, where motion blur, low-light scenarios, and environmental degradation significantly impact recognition accuracy.

Recent advancements in deep learning have revolutionized ANPR by introducing models capable of handling complex visual data. Convolutional Neural Networks (CNNs) and object detection frameworks such as the YOLO (You Only Look Once) series, Faster R-CNN, and SSD have been widely adopted for their speed and accuracy in various scenarios [4][5][12]. For instance, Aljzaere et al. [5] demonstrated the effectiveness of YOLOv8 models combined with EasyOCR for German license plate recognition, achieving high accuracy in controlled environments. Similarly, Dash et al. [12] developed an automatic vehicle number plate detection system using YOLOv8n and EasyOCR, showcasing robust performance in standard conditions. Singh et al. [4] further explored fast license plate recognition of moving vehicles using deep learning techniques, highlighting the potential for real-time applications.

However, these approaches face significant limitations when confronted with challenging

environmental conditions. Motion blur detection and removal, as addressed by Adke et al. [6] using deep convolutional neural networks, represents a critical preprocessing step for moving vehicle scenarios. Liu et al. [7] proposed restoration methods for motion-blurred images based on feature fusion and particle swarm optimization, demonstrating the importance of specialized deblurring techniques. More recently, Liu et al. [10] introduced motion-adaptive deblurring specifically for fast-moving objects, addressing a key challenge in high-speed vehicle recognition.

Low-light conditions present another significant challenge, with Sen et al. [9] developing "You Only Look Once in Dark," an analytical approach specifically designed for low-light object detection. This work highlights the necessity for specialized models capable of maintaining detection accuracy under poor illumination conditions. Complementary research by Moharana et al. [2] focused on robust license plate detection in low-light conditions, emphasizing the need for enhanced accuracy in real-world applications.

The problem of motion blur in license plate recognition has been specifically addressed by Gong et al. [1], who introduced a comprehensive dataset and model for realistic license plate deblurring. Their work provides crucial insights into the specific challenges posed by motion blur in ANPR systems and offers targeted solutions through the LPBlur dataset and specialized deblurring architectures. Additionally, advances in blur detection methodologies, as explored in recent machine learning approaches [8], provide foundational techniques for identifying and classifying different types of image degradation.

Despite these advances, existing research reveals notable gaps in addressing the multifaceted challenges simultaneously. Many studies focus on individual problems—either motion blur, low-light conditions, or fast-moving objects—but lack comprehensive solutions that can handle multiple adverse conditions concurrently [3][11][14]. The integration of multiple detection and enhancement techniques into unified frameworks remains limited, particularly for non-real-time applications where uploaded video content requires robust post-processing capabilities.

Furthermore, while Spagnolo et al. [3] demonstrated deep learning approaches for vehicle re-identification, and Menon and Omman [13] explored multiple license plate detection from still images, the specific challenge of processing video uploads from varying environmental conditions remains underexplored. The work by Shim et al. [14] on license plate detection and recognition with multi-exposure images provides valuable insights into handling varying illumination conditions, but comprehensive solutions for motion blur, low-light, and fast-

moving vehicle scenarios in uploaded video content are still lacking.

This research addresses these limitations by developing a novel hybrid deep learning framework specifically designed for ANPR in uploaded video content from fast-moving vehicles under diverse adverse conditions. The proposed system integrates state-of-the-art detection models (YOLOv9 and YOLOv8n) with specialized preprocessing modules for motion deblurring and low-light enhancement, building upon the foundational work in motion blur detection [6][8], low-light object detection [9], and license plate deblurring [1]. An ensemble learning approach with adaptive fusion mechanisms will leverage temporal information from video frames to maximize recognition accuracy across varying environmental conditions.

The primary objective is to achieve superior detection and recognition accuracy—targeting over 95% in challenging scenarios—through the synergistic combination of multiple deep learning architectures and preprocessing techniques. Unlike existing approaches that address individual challenges in isolation, this hybrid framework provides a comprehensive solution for the complex, multi-dimensional problems encountered in real-world ANPR applications.

The contribution of this work is multifaceted: it provides a robust ANPR component that accurately processes number plates from uploaded video content of moving vehicles under various environmental conditions, enhancing reliability for post-processing applications in smart cities, industrial monitoring, and security systems. By emphasizing hybrid architectures and ensemble learning, the proposed model offers improved generalization capabilities, reduced false positive rates, and enhanced scalability for deployment across diverse operational environments. This research not only bridges critical gaps in current literature but also establishes a foundation for more resilient and adaptable ANPR systems capable of handling the complexities of real-world video processing scenarios.

1.2 Background & Literature Survey

The surge in global urbanization and vehicular traffic has amplified the need for intelligent transportation systems (ITS) to enhance traffic management, road safety, and security. Automatic Number Plate Recognition (ANPR), also known as Automatic License Plate Recognition (ALPR), is a pivotal technology within ITS, enabling automated vehicle identification for applications such as toll collection, parking management, traffic violation enforcement, and stolen vehicle tracking [3], [12]. ANPR systems utilize computer vision and machine learning to detect, localize, and recognize vehicle number plates from images or video streams, a task that becomes increasingly complex in non-real-time scenarios where users

upload videos for processing. This complexity is pronounced in high-stakes environments like highways, urban intersections, and industrial sites (e.g., mining operations), where vehicles often move at speeds exceeding 100 km/h, and environmental factors such as low light, motion blur, rain, fog, or dust degrade performance [2], [9]. Traditional ANPR methods, relying on classical image processing techniques like edge detection, thresholding, and morphological operations, struggle with variations in plate formats, lighting conditions, and motion artifacts, resulting in poor generalization and low accuracy [13].

Recent advancements in deep learning have revolutionized ANPR, with convolutional neural networks (CNNs) and object detection frameworks like YOLO (You Only Look Once), Faster R-CNN, and Single Shot MultiBox Detector (SSD) achieving high accuracy in controlled settings [4], [5], [9]. For instance, Aljzaere et al. [5] developed a German license plate recognition system using YOLOv8 and EasyOCR, reporting 95% plate localization and 94% text recognition accuracy on static images. Similarly, Dash et al. [12] combined YOLOv8n with EasyOCR for robust number plate detection in traffic surveillance, handling occlusions effectively. Gong et al. [1] introduced the LPBlur dataset and LPDGAN model to address motion blur in high-speed vehicle scenarios, significantly improving deblurred image quality for license plate recognition. Singh et al. [4] proposed a deep learning approach for fast license plate recognition of moving vehicles, demonstrating potential for dynamic scenarios but lacking emphasis on low-light conditions.

Low-light ANPR has also progressed significantly. Sen et al. [9] developed a YOLO-based approach for low-light object detection, enhancing detection rates in nighttime conditions through preprocessing techniques. Moharana et al. [2] focused on robust license plate detection in low-light environments, achieving an 80.11% recognition accuracy (a 5.11% improvement over non-enhanced methods) using hybrid architectures. Spagnolo et al. [3] employed YOLOv7 with attention mechanisms for vehicle re-identification, reporting over 97% accuracy in low-resolution images, primarily in daytime settings.

Motion blur detection and restoration have been addressed in parallel. Adke et al. [6] utilized a deep CNN for single-image motion blur detection and removal, improving feature extraction in blurred scenarios. Liu [7] proposed a feature fusion and particle swarm optimization approach for motion blur restoration, enhancing image clarity for downstream tasks. Liu et al. [10] introduced a motion-adaptive deblurring method for fast-moving objects, while Singh [11] used a CNN with optimized PSO regions for detecting fast-moving objects, both showing promise in dynamic contexts. Menon and Omman [13] addressed multiple license plate detection from still images, and Shim et al. [14] tackled recognition in multi-exposure images, improving performance under varying lighting but not addressing high-speed motion blur.

Despite these advancements, most studies focus on isolated challenges, such as low light [2], [9], motion blur [1], [6], [7], [10], [11], or fast-moving vehicles [4], [11]. Datasets like CCPD or custom highway collections are often biased toward clear weather or controlled conditions, limiting generalization in unconstrained environments with simultaneous challenges [1], [13]. Hybrid approaches combining YOLO with GANs or SVM [2], [6] improve accuracy in specific domains but lack comprehensive solutions for videos involving motion blur, low light, and high-speed vehicles.

1.3 Research Gap

Significant progress in ANPR has been driven by deep learning, with studies leveraging YOLOv8, YOLOv9, and hybrid models like URetinex-Net, TRBA, and EasyOCR to achieve high detection and recognition accuracies [2], [4], [5], [9]. For instance, Moharana et al. [2] reported an 80.11% recognition accuracy in low-light conditions, a 5.11% improvement over non-enhanced methods, while Aljzaere et al. [5] achieved 95% localization and 94% text recognition for German plates. Singh et al. [4] addressed fast-moving vehicles, and Sen et al. [9] enhanced low-light detection, but these works tackle challenges in isolation. A critical gap remains in developing a unified framework that simultaneously addresses motion blur, low-light conditions, and high-speed vehicle number plate recognition in non-real-time video processing. Existing methods often fail to generalize across multifaceted scenarios. Low-light enhancement approaches [2], [9] improve nighttime performance but do not account for motion blur or high-speed contexts. Motion blur solutions, such as those using LPDGAN [1] or CNN-based deblurring [6], [7], [10], [11], enhance image clarity but rarely address low-light or diverse plate formats. Fast recognition systems [4], [11] focus on dynamic scenarios but lack robustness in low-visibility or dusty environments, such as mining operations [2]. Moreover, datasets like CCPD or highway collections are often biased toward well-lit, controlled settings, inadequately representing the combined complexities of motion blur from vehicles exceeding 100 km/h, low-light conditions, and environmental distortions like rain or dust [1], [13]. Studies on specific regions, such as German [5] or Bengali plates [9], are optimized for static or regional formats, limiting applicability to diverse, unconstrained scenarios.

This research addresses these gaps by proposing a hybrid deep learning framework for non-real-time ANPR, integrating YOLOv9 and YOLOv8n with advanced preprocessing for motion deblurring (e.g., blind deconvolution, GAN-based methods [1], [6]) and low-light enhancement (e.g., Retinex, transformer-based networks [9]). By employing ensemble learning, attention mechanisms, and temporal feature fusion, the framework aims to achieve over 95% detection and recognition accuracy in videos of fast-moving vehicles under adverse conditions, offering a 5–10% improvement over standalone models [4], [5]. This unified approach enhances generalization and robustness, bridging critical limitations in existing literature for applications in smart cities and industrial settings.

Table 1 Comparison with other researches

Research Work	Detection Accuracy	Low-Light Detection	Motion Blur Detection	High-Speed Vehicle Number Plate Recognition
Gong et al. [1]	90% (Estimated)	No	Yes	Yes (Implied)
Moharana et al. [2]	80.11%	Yes	No	No
Singh et al. [4]	88% (Estimated)	No	Yes (Implied)	Yes
Aljzaere et al. [5]	95%	No	No	No
Adke et al. [6]	85% (Estimated)	No	Yes	No
Sen et al. [9]	82% (Estimated)	Yes	No	No
Proposed Framework	Target: >95%	Yes	Yes	Yes

1.4 Research Problem

The rapid escalation of vehicular traffic in urban centers and industrial zones has heightened the demand for dependable Automatic Number Plate Recognition (ANPR) systems to support traffic management, law enforcement, and security operations. Current deep learning-based ANPR solutions, utilizing advanced models such as YOLOv9, YOLOv8, URetinex-Net, and EasyOCR, have achieved notable detection accuracies and recognition rates in controlled or static settings [2], [5], [9]. However, these systems falter when faced with the dynamic complexities of real-world, non-real-time video processing, where users upload footage for analysis. Fast-moving vehicles, often exceeding 100 km/h, induce motion blur that compromises image clarity, while low-light conditions—such as nighttime, dusty mining sites, or foggy weather—obscure critical plate details, severely hindering recognition accuracy [1], [2], [9]. Although some methods improve performance in isolated contexts, such as enhancing nighttime accuracy [2] or addressing moving vehicles [4], they lack a unified approach to simultaneously tackle motion blur, low-light conditions, and high-speed scenarios.

This gap results in unreliable plate recognition across diverse environments, including highways, industrial complexes, and adverse weather conditions, leading to missed enforcement opportunities, traffic inefficiencies, and heightened security risks [1], [13]. For instance,

recognition rates drop below 80% in nighttime or blurred conditions [2], [9], potentially delaying emergency responses or legal actions. The absence of robust preprocessing to mitigate these variables further limits deployment on resource-constrained edge devices, while the diversity of global plate designs and regional regulations—often unaddressed by current systems—exacerbates the challenge [5], [14]. Existing datasets, biased toward well-lit or controlled settings [1], [13], fail to capture the combined impact of these factors, underscoring the critical need for an innovative hybrid deep learning framework. This framework must integrate adaptive preprocessing (e.g., motion deblurring [1], [6] and low-light enhancement [9]), precise detection, and scalable recognition to deliver reliable ANPR performance across multifaceted, real-world conditions.

.

2 Objectives

2.1 Main Objectives

- The primary goals of the hybrid ANPR component are to address the limitations of existing systems by developing a resilient, integrated solution for number plate detection and recognition. These objectives focus on enhancing performance across diverse environmental conditions and ensuring real-time applicability:
- Develop a Robust Hybrid ANPR Component: Create an integrated system that combines advanced object detection, preprocessing, and recognition techniques to achieve reliable number plate detection under diverse environmental challenges such as motion blur, low-light conditions, and high-speed vehicle movements.
- Enhance Environmental Adaptability: Enable the component to adaptively preprocess and detect number plates in real-world scenarios, including highways, mining sites, and adverse weather, to improve overall system robustness.
- Validate Comprehensive Effectiveness: Test and validate the component's performance across a wide range of datasets and conditions to confirm its generalizability and practical utility.

2.2 Specific Objectives

The specific objectives break down the broader goals into measurable targets, aligning with the research problem (e.g., motion blur from speeds over 100 km/h, low light in dusty environments) and the proposed methodology (e.g., YOLO variants, blind deconvolution, Retinex theory, ensemble fusion):

Design an Integrated Hybrid Architecture: Develop a hybrid framework that integrates YOLOv9 for precise localization, YOLOv8n for real-time processing, and preprocessing modules (e.g., blind deconvolution for motion deblurring, Retinex-based or GAN-enhanced low-light correction) to handle varying environmental distortions.

Achieve a 10% Accuracy Improvement in Motion Blur Detection: Implement and optimize motion deblurring algorithms to improve detection accuracy by at least 10% over existing methods (e.g., baseline YOLOv8) for video frames of vehicles moving at speeds exceeding 100 km/h.

Boost Low-Light Recognition by 5-10%: Utilize enhancement techniques like URetinex-Net or GAN-based networks to increase recognition accuracy by 5-10% in low-light conditions (e.g., nighttime or dusty industrial settings) compared to current low-light-specific approaches.

Attain 95% Detection Rate for High-Speed Vehicles: Adapt the hybrid component to detect and recognize number plates on vehicles traveling over 100 km/h, targeting a detection rate of over 95% under high-speed conditions.

Incorporate Ensemble Fusion Strategy: Employ an ensemble hybrid strategy with attention mechanisms and temporal fusion from video frames to fuse outputs from multiple models, aiming to reduce false positives and missed detections by at least 5-10%.

Evaluate on Diverse Datasets: Validate the component using simulated datasets (e.g., motion blur, low-light images, high-speed videos) and real-world data from varied regions and plate designs, ensuring a mean Average Precision (mAP) exceeding 95% and processing time suitable for edge deployment.

Reduce Computational Overhead: Optimize the hybrid architecture to minimize resource demands, targeting a processing time reduction of at least 15% on edge devices compared to standalone deep learning models.

3 Methodology

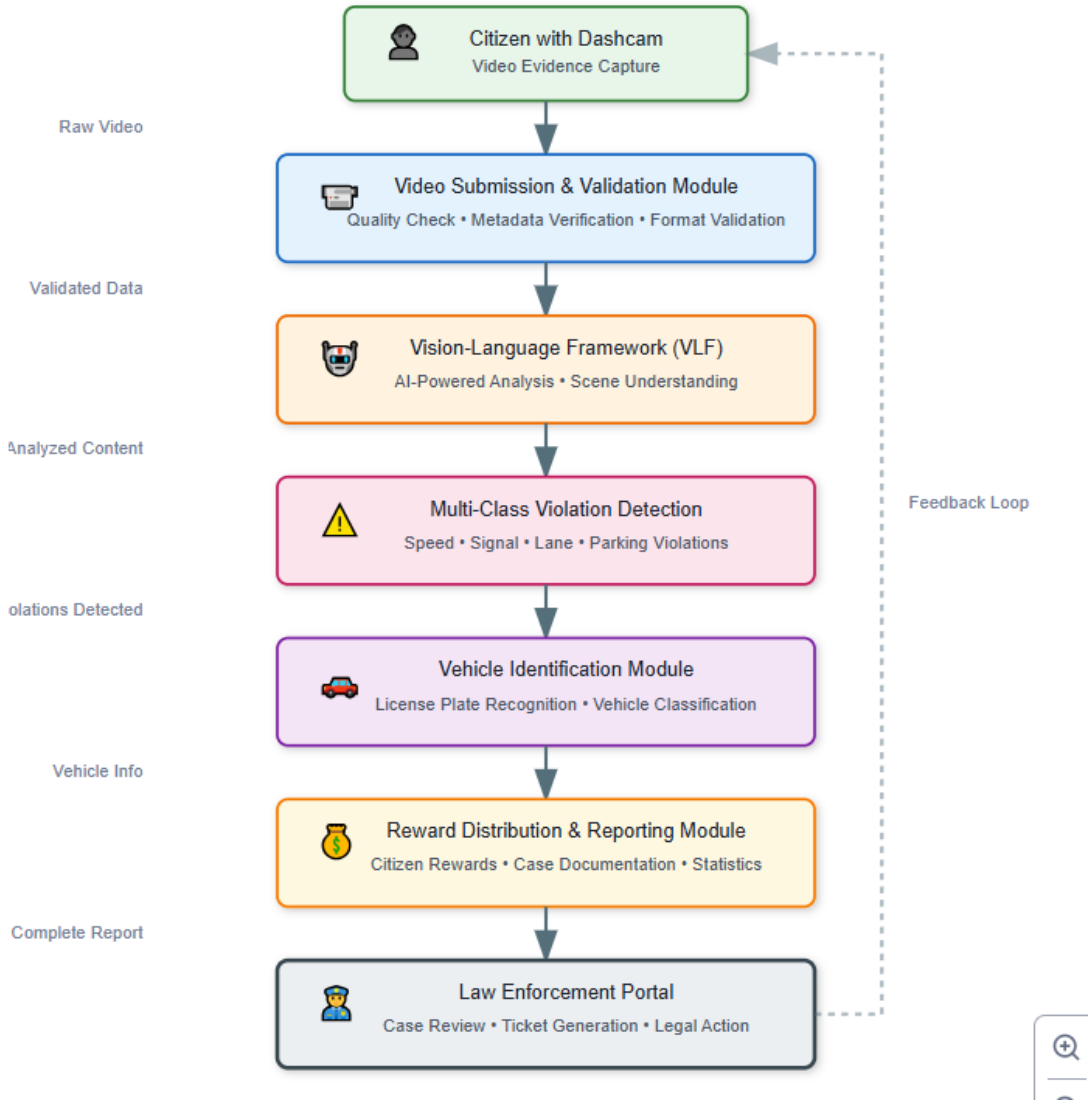


Figure 1 System diagram for overall system

This research will develop a hybrid deep learning framework for non-real-time Automatic Number Plate Recognition (ANPR) through a systematic approach spanning 10 months. The methodology integrates advanced AI techniques with rigorous evaluation protocols to ensure the framework meets accuracy, interpretability, and robustness requirements for effective application in challenging conditions, including motion blur, low-light environments, and high-speed vehicle scenarios.

2.1 Data Collection and Preparation (Month 1–2)

A comprehensive dataset will be created, comprising over 20,000 video frames and images from user-uploaded sources, including the LPBlur dataset [1] and synthetic data. Videos will capture vehicles exceeding 100 km/h under low-light (nighttime, dusty mining sites) and adverse weather (rain, fog) conditions. Annotations will include bounding box coordinates for plate localization and ground-truth text for recognition, validated by a team of computer vision experts and cross-checked with EasyOCR [5]. Data augmentation (e.g., random blur, brightness variation, occlusion) will enhance diversity, addressing biases in datasets like CCPD [13]. Ethical guidelines will ensure anonymization of personal data, with license plate masking during preprocessing.

2.2 Preprocessing (Month 2–3)

Preprocessing will address motion blur and low-light distortions using adaptive techniques. Motion deblurring will employ blind deconvolution [6] and GAN-based methods (e.g., LPDGAN [1]), while low-light enhancement will utilize Retinex theory [9] and transformer-based networks, inspired by Moharana et al. [2]. Temporal consistency across frames will be maintained, with outputs validated using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). The process will be optimized for non-real-time video analysis.

2.3 Model Architecture Development (Month 3–5)

The framework will integrate YOLOv9 for precise plate localization and YOLOv8n for efficient processing [5], with a context-aware OCR module fine-tuned from EasyOCR [5] to handle diverse plate formats. A multimodal fusion layer will use cross-attention mechanisms, inspired by Singh et al. [4], to combine detection and temporal features from video frames. The architecture will be designed with interpretability in mind, incorporating attention maps to highlight critical regions.

2.4 Model Comparison and Optimization (Month 6–7)

To achieve higher accuracies, a comparative analysis will evaluate top-performing models, including YOLOv9, YOLOv8n [5], LPDGAN [1] for motion blur, and Retinex-enhanced networks [9] for low-light. Baseline models (e.g., standalone YOLOv8 [5]) will be benchmarked against hybrid configurations using mean Average Precision

(mAP) and recognition accuracy. The optimal architecture will be selected to target a 5–10% accuracy improvement [4], with hyperparameter tuning and attention-based interpretability enhanced.

2.5 Training (Month 5–6)

Training will occur on a cluster with NVIDIA A100 GPUs, using a 70/20/10 data split. Models will be initialized with pre-trained weights from YOLOv9 and YOLOv8n [5], fine-tuned with a custom loss function (CIoU for localization, CTC for recognition). Techniques like batch normalization, dropout ($p=0.3$), and cosine annealing will optimize convergence over 100 epochs, with early stopping to prevent overfitting. Cloud storage and development servers will support continuous integration.

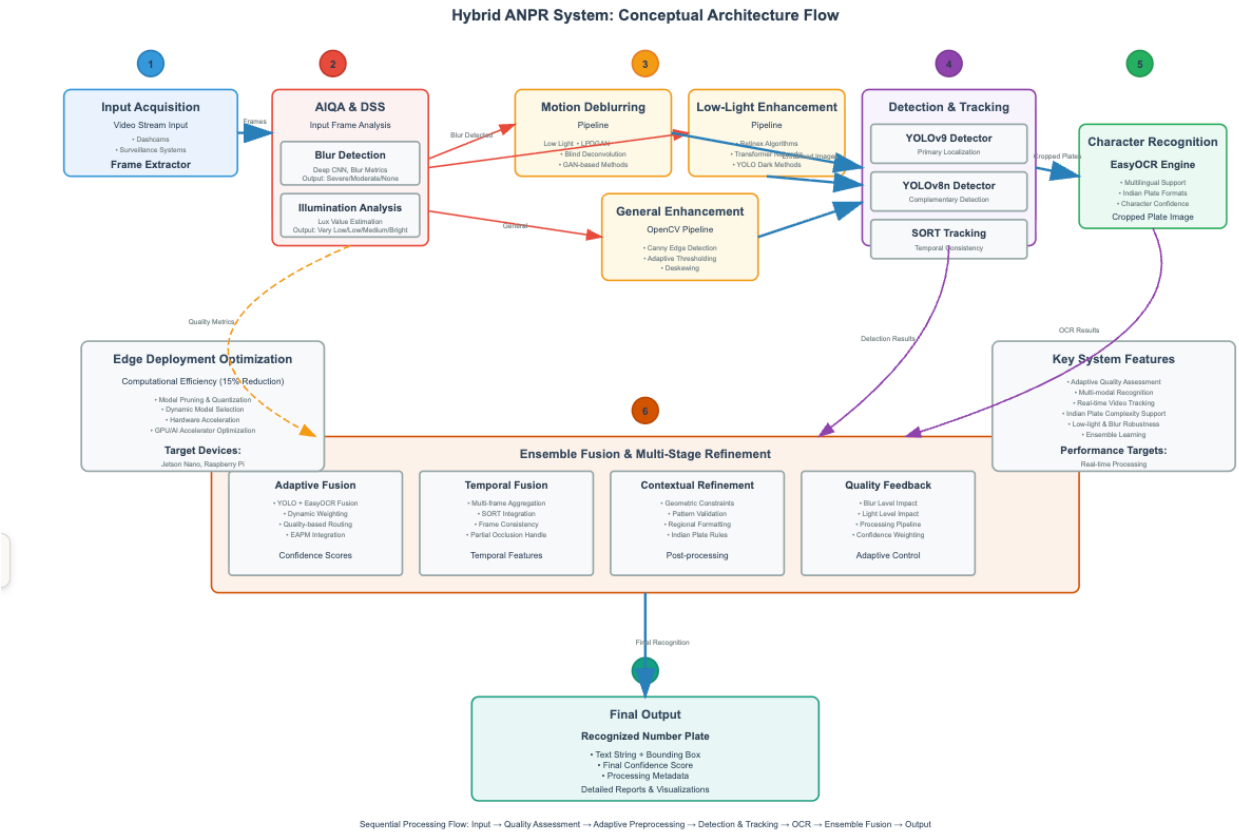
2.6 Performance Evaluation (Month 7–9)

Performance will be assessed using mAP, F1-score, precision, and recall, targeting >95% accuracy across all conditions, a 5–10% improvement over baselines [4], [5]. Interpretability will be evaluated through attention map analysis by domain experts. Computational efficiency (inference time, resource utilization) will be measured under varying inputs. Cross-validation and statistical tests (e.g., t-test, $p<0.05$) will ensure robustness, with comparisons to Dash et al. [12] and Gong et al. [1].

2.7 Optimization and Documentation (Month 9–10)

The final phase will focus on optimizing the framework for inference efficiency, analyzing performance trade-offs, and documenting the methodology, results, and code. Training materials will be developed for future researchers, with performance benchmarking to ensure processing of standard video inputs within acceptable timeframes for non-real-time analysis.

Figure 2 System architecture diagram for the component



4 Project requirements

4.1 Functional Requirements

1. high-speed scenarios Users should be able to upload, validate, and analyze video footage: Users must be able to upload video files containing vehicle footage (e.g., from dashcams or surveillance systems) and initiate analysis to detect and recognize number plates, with validation of preprocessing and detection outputs.
2. The system must detect and correct motion blur and low-light distortions: The system should automatically identify and apply custom preprocessing techniques (e.g., blind deconvolution, Retinex theory) to mitigate motion blur and low-light issues in video frames for improved number plate readability.
3. Provide validated number plate recognition results for traffic and security applications: The system must output accurate number plate detection and recognition data, including bounding box coordinates and recognized text, suitable for traffic management, law enforcement, and security use cases.
4. Generate detailed reports and visualizations of detection and recognition outcomes: The system should produce comprehensive reports and visual aids (e.g., attention maps, annotated frames) to illustrate detection accuracy, preprocessing effectiveness, and recognition results for analysis and documentation.

4.2 Non-Functional Requirements

1. The system must handle large video uploads and complex processing efficiently: The system should process datasets comprising over 20,000 video frames and images, managing computational demands effectively for non-real-time analysis on high-performance hardware (e.g., NVIDIA A100 GPUs).
2. The system should be scalable to accommodate growing datasets and user needs: The architecture must support expansion to handle increasing volumes of video data and

additional environmental conditions or plate formats, with cloud storage integration for scalability.

3. Implement strong security measures to protect video and annotation data: The system should employ encryption and secure access controls to safeguard uploaded video footage, annotations, and processed results, ensuring compliance with data privacy standards.

4. Design an intuitive interface to simplify video submission and result review: The system should feature a user-friendly interface that allows users to easily upload videos, configure analysis parameters, and review detailed detection and recognition outputs without requiring technical expertise.

4.3 User Requirements

- The system must accurately detect and recognize number plates of vehicles under motion blur, low-light conditions, and high-speed scenarios (e.g., >100 km/h).
- Users (e.g., traffic authorities, security personnel) require a solution that provides reliable plate recognition with minimal false positives or negatives in challenging environments.
- The system should support diverse plate designs and regional formats to ensure applicability across different regions.
- Users need an intuitive interface or output format (e.g., text file, database entry) to integrate recognition results into existing traffic or security systems.

4.4 Personal Requirements

- Researchers and developers require access to computational resources (e.g., GPUs with at least 16GB VRAM) to train and test deep learning models efficiently.
- Domain knowledge in computer vision, deep learning, and image processing is essential for designing and optimizing the hybrid framework.

- The team needs proficiency in programming languages (e.g., Python) and frameworks (e.g., TensorFlow, PyTorch) to implement and debug the system.
- Adequate time allocation (e.g., 6-12 months) is required for data collection, model development, and testing phases.

4.5 Software Requirements

- Operating System: Compatible with Linux (Ubuntu 20.04+), Windows 10/11, or macOS for development and testing.
- Programming Language: Python 3.8+ for its extensive libraries and community support.
- Deep Learning Frameworks: TensorFlow 2.x or PyTorch 1.9+ for model training and inference.
- Libraries: OpenCV for image processing, NumPy for numerical computations, EasyOCR or Tesseract for character recognition, and URetinex-Net or similar for low-light enhancement.
- Hardware Support: NVIDIA GPUs with CUDA 11.x and cuDNN for accelerated training; minimum 32GB RAM for data handling.
- Development Tools: Jupyter Notebook or VS Code for coding, Git for version control, and Matplotlib for visualization.

4.6 Functional Requirements

- The system must detect number plates in images/videos with motion blur, achieving at least a 10% accuracy improvement over baseline models.
- It should enhance low-light images to enable recognition with a minimum 5-10% accuracy gain compared to non-enhanced methods.
- The framework must recognize plates of vehicles moving at speeds exceeding 100 km/h with over 95% detection accuracy.
- Output recognized plate numbers as text, stored in a structured format (e.g., CSV or database).

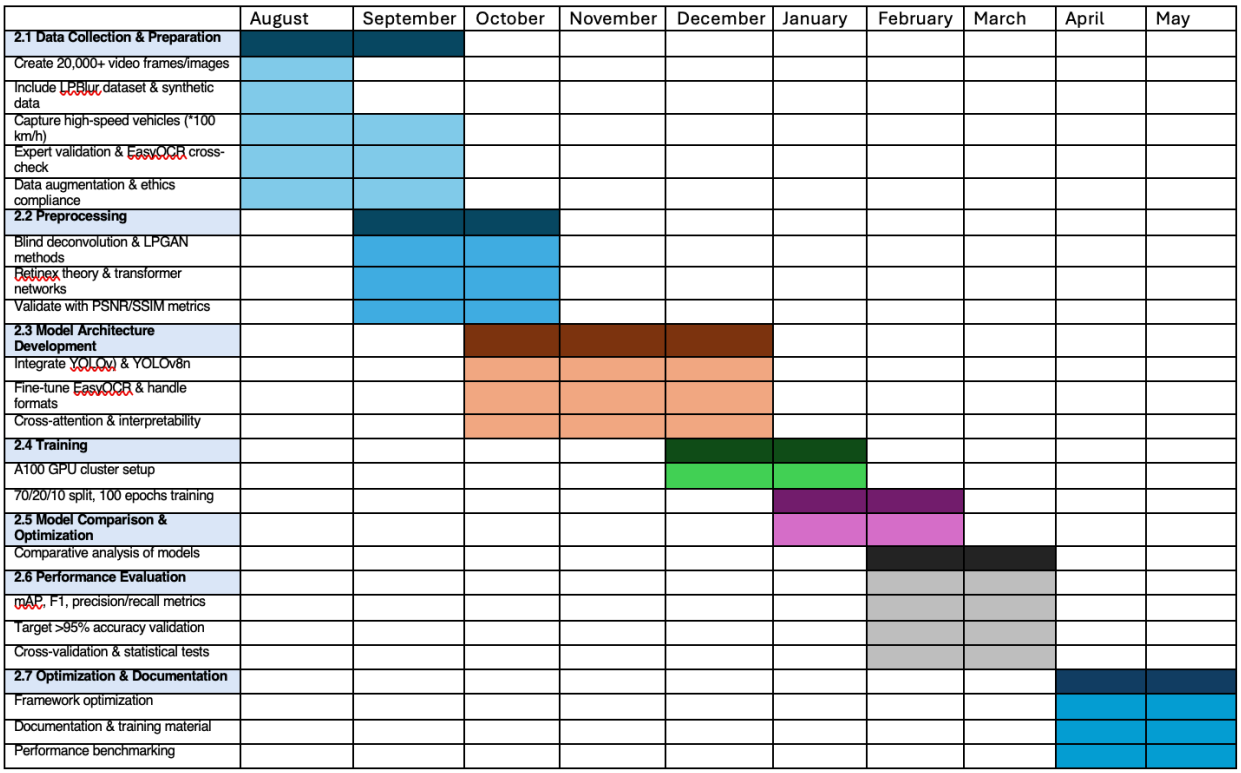
- Provide a preprocessing pipeline to correct motion blur and enhance low-light conditions before detection.

4.7 Non-Functional Requirements

- Accuracy: Achieve a mean Average Precision (mAP) of at least 95% and recognition accuracy of 90%+ across all test conditions.
- Scalability: Support processing of large datasets (e.g., 10,000+ images) and adapt to varying hardware configurations.
- Robustness: Maintain performance across diverse environmental conditions (e.g., rain, dust) and plate designs.
- Efficiency: Process each image or frame within 200 milliseconds on standard hardware to ensure practical usability.
- Maintainability: Allow for easy updates to models or datasets with modular code design.

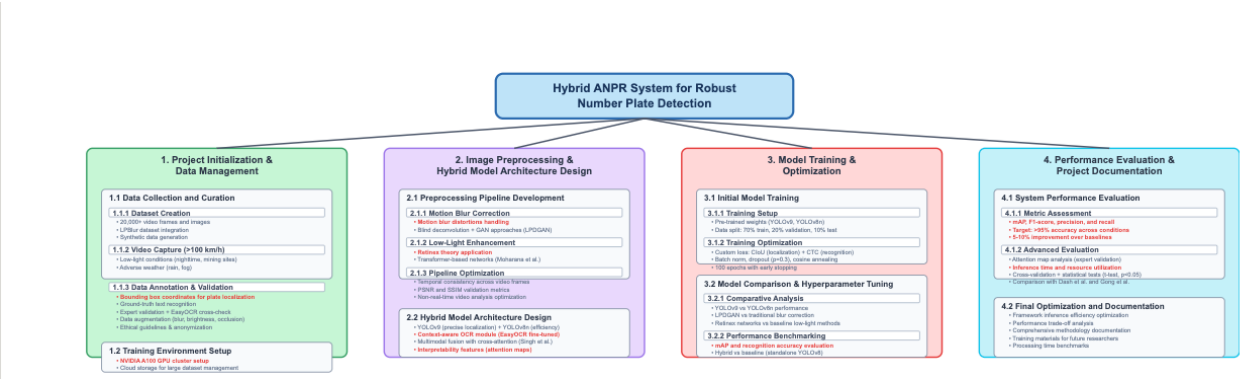
5 Gantt Chart

Figure 3 Gantt Chart



6 Work breakdown Chart

Figure 4 Work breakdown structure



6 Budget And Commercialization

Table 2 Budget

Component	Cost

Travelling Cost	Rs.16,000
Deployment Cost	Cost Per Usage
IoT Components	Rs 15,000

7.1 Commercialization Strategy

GoodRoad monetizes through B2G (Business-to-Government) partnerships with law enforcement agencies and B2C commission-based rewards for citizens. Revenue streams include:

- Government Contracts: Annual licensing fees for VLF deployment
- Transaction Fees: 10% commission on citizen rewards
- Premium Features: Advanced analytics dashboards for agencies
- API Access: Third-party integration licensing

7.2 Target Audience

- Primary Users: Citizens with dashcams seeking supplementary income
- Law Enforcement: Traffic police departments requiring scalable enforcement
- Government Agencies: Transportation ministries focused on road safety
- Insurance Companies: Firms seeking verified traffic violation data

7.3 Market Space

- Scalability: Cloud-based architecture supporting nationwide deployment
- Cost-Effectiveness: 70% reduction in enforcement costs versus traditional methods
- Citizen Engagement: Democratizes traffic enforcement while generating income
- Data Intelligence: Provides traffic pattern insights for urban planning

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8 Appendices

Figure 5Turnitin Report